

Cryptography

“kryptos” – hidden, secret

“graphein” – writing

Cryptography has to do with making information hidden

Cryptanalysis: analysis of that which is hidden (i.e., for the purpose of understanding/uncovering/revealing it)

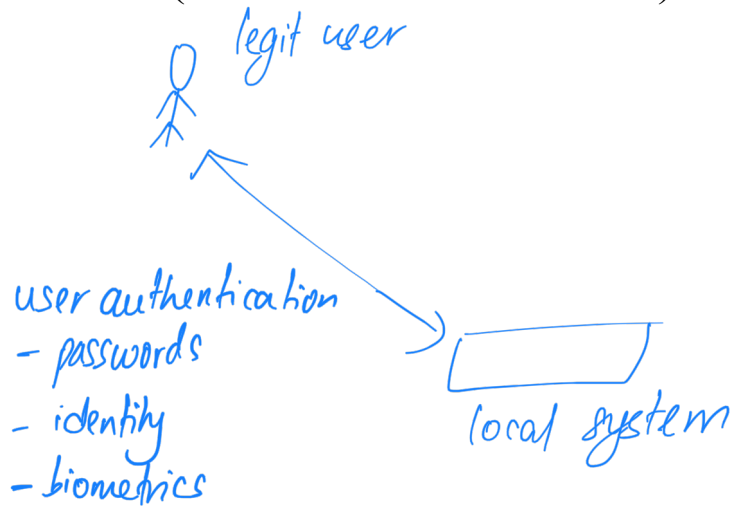
Cryptology: the study of hiding (information) – making and breaking ciphers

Historical definition to modern definition [Katz & Lindell]:

- Art form that deals with secret communication for the military and intelligence communities
- A science that helps to secure systems for ordinary people across the globe

[*SEE COURSE OUTLINE***]**

Setting the context: designing a secure computer system
(see CSI4139 / CEG4399)



Historical perspective:

Cryptography has probably been around as long as language has existed (as long as people have communicated, there has been a desire to hide some communications from other parties)

At a minimum, cryptography has a history of at least 4000 years [Kranakis]

- The ancient Egyptians enciphered some of their hieroglyphic writing on monuments
- The clay of Phaistos (Cretan-Minoan, 17th century BC) has still not been deciphered
- The ancient Hebrews enciphered certain words in the scriptures
- Herodotus (ca. 430 BC) describes how encrypted messages were transported by messengers
- Julius Caesar (ca. 50 BC) enciphered his correspondence in personal and military affairs
- Roger Bacon described several enciphering methods in the 1200s
- Geoffrey Chaucer (1300s) included several ciphers in his works
- Leon Alberti devised a cipher wheel and described the principles of frequency analysis in the 1460s
- Blaise de Vigenere published a book on cryptology in 1585; described the polyalphabetic substitution cipher

(See *The Codebreakers*, by David Khan, for many, many more examples.)

Basic concepts

Some possible goals in this scenario:

Fundamental properties of an encryption algorithm:

- An encryption scheme consists of a *key generation algorithm* G (which generates keys in a valid key space K), an *encryption algorithm* E (which encrypts messages from a valid message space M to a ciphertext space C), and a *decryption algorithm* D (which decrypts messages from C to M). For every k, k' output by G and every $m \in M$, it holds that

$$D_{k'}(E_k(m)) = m$$

- The encryption and decryption functions are efficiently computable for all keys (i.e., it should be relatively easy both to encrypt and to decrypt, given the key)
- It should be *hard* for an attacker to determine the unknown information (e.g., the original plaintext string, or the key k that was used) from the known information

“Hard” ???

Model of Eve (the attacker): *The attacker can know what was the encryption algorithm.*

Kerckhoffs' Principle:

- "The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience."

I.e., security must **rely solely on the secrecy of the key**

Why? *Easier to keep a short key secret than a big algorithm secret*

- Secrecy of short key versus big algorithm
- Simplicity of changing key versus replacing algorithm *easier to change key than the algo.*
- For ^{large} groups: one alg. w/ many keys vs. many algs. *easier*

Today we go further: we don't just assume that the algorithm might not be secret, but we demand that it be made public

- Published designs **undergo public scrutiny**
- Better if **flaws** are discovered by **"ethical hackers"** than by malicious parties
- **Protects** society from malicious designers *→ deliberately design a weak algorithm*
- Enables the establishment of standards *→ can have standardised algo. Obtain interoperability among different countries, parties, departments that can communicate with each other.*

Eve's ability:

- Ciphertext-only attack *(Simplest : knows nothing but the ciphertext).*
- Known-plaintext attack *(already knows some of the plaintext)*
- Chosen-plaintext attack *(Eve can influence Alice & give her some plaintext to encrypt)*
- Chosen-ciphertext attack *(Eve can influence Bob & give him some ciphertext to decrypt).*
- Related key attack *(Alice & Bob are using the same key & can influence both parties to change their keys)*

increasingly powerful for the attacker

keep data hidden? → Is the message unchanged? → knowing where the message came from

Goals of a system designer that wants to use cryptography:

- Confidentiality, Integrity, Authenticity, Cryptographic Strength

↳ measure strength of algo.

Cryptographic strength:

There is only 1 algorithm that has this property: ONE TIME PAD

- **Unconditional security** – no matter how much computer power or time is available, the cipher cannot be broken. The ciphertext does not have enough information to uniquely determine the plaintext (i.e. many plaintext values could have given this ciphertext).
- **Computational security** – given **limited resources** (e.g., time, money, computing power), the cipher cannot be broken

Exhaustive search attack:

Simplest possible measure of strength: given a ciphertext, try every possible key (“brute force attack”)

- Assumes corresponding plaintext is either known or can be recognized

32-bit key is not secure.
56-bit key is not secure either.
minimum key size: 128 bits

(See Stallings, Table 2.2.)

Of course other, more effective, attacks are often possible for any given cipher...

Analyze the flaws of an algo.